

Companion to Bird & de Moor, Algebra of Programming

§0. Laws from Relation Algebra

§0.1. Adjunctions

(0.1a)

	$F \dashv G$	F's type	monad FG	$1 \sqsubseteq FG$	$GF \sqsubseteq 1$	F preserves $\cup$	G preserves $\cap$	$FGF=F$	$GFG=G$
1	$\triangleleft \dashv \triangleright$	$A \rightarrow A \otimes A$	$\triangleleft \triangleright = 1$	$1 \sqsubseteq \triangleleft \triangleright$	$\triangleright \triangleleft \sqsubseteq 1$	$(R \cup S) \triangleleft = R \triangleleft \cup S \triangleleft$	$(R \cap S) \triangleright = R \triangleright \cap S \triangleright$	$\triangleleft \triangleright \triangleleft = \triangleleft$	$\triangleright \triangleleft \triangleright = \triangleright$
2	$\multimap \dashv \multimap$	$A \rightarrow I$	$\multimap \multimap = T$	$1 \sqsubseteq \multimap \multimap$	$\multimap \multimap \sqsubseteq 1$	$(R \cup S) \multimap = R \multimap \cup S \multimap$	$(R \cap S) \multimap = R \multimap \cap S \multimap$	$\multimap \multimap \multimap = \multimap$	$\multimap \multimap \multimap = \multimap$
3	$\circ \dashv \circ$	$(A \rightarrow B) \rightarrow (B \rightarrow A)$	$R^{\circ \circ} = R$	$R^{\circ \circ} = R$	$R^{\circ \circ} = R$	$(R \cup S)^{\circ} = S^{\circ} \cup R^{\circ}$	$(R \cap S)^{\circ} = R^{\circ} \cap S^{\circ}$	$R^{\circ \circ} = R$	$R^{\circ \circ} = R$
4	$\multimap \triangleleft \triangleright \multimap$	$(X \otimes A \rightarrow Y) \rightarrow (X \rightarrow Y \otimes A)$	1	$(\multimap \triangleleft \otimes 1)$ $(1 \otimes \triangleright \multimap) = 1$	$(1 \otimes \multimap \triangleleft)$ $(\triangleright \multimap \otimes 1) = 1$	$=$	$=$	$=$	$=$
5	$\triangleright \multimap \triangleleft \triangleright$	$(X \rightarrow Y \otimes A) \rightarrow (X \otimes A \rightarrow Y)$	1	$(1 \otimes \multimap \triangleleft)$ $(\triangleright \multimap \otimes 1) = 1$	$(\multimap \triangleleft \otimes 1)$ $(1 \otimes \triangleright \multimap) = 1$	$=$	$=$	$=$	$=$
6	$\Delta \dashv \cap$	$(A \rightarrow B) \rightarrow (A \rightarrow B)^2$	$R \cap R = R$	$R \cap R = R$	$R \cap S \sqsubseteq R$	$\Delta(R \cup S) = \Delta R \cup \Delta S$	$(R \cap T) \cap (S \cap U) = (R \cap S) \cap (T \cap U)$	$R \cap R = R$	$R \cap R = R$
7	$\cup \dashv \Delta$	$(A \rightarrow B)^2 \rightarrow (A \rightarrow B)$	$(R, S) \mapsto (R \cup S, R \cup S)$	$R \sqsubseteq R \cup S$	$R \cup R = R$	$(R \cup T) \cup (S \cup U) = (R \cup S) \cup (T \cup U)$	$\Delta(R \cap S) = \Delta R \cap \Delta S$	$R \cup R = R$	$R \cup R = R$
8	$\emptyset \dashv !$	$\{*\} \rightarrow (A \rightarrow B)$			$\emptyset \sqsubseteq R$				
9	$\cdot S \dashv / S$	$(A \rightarrow B) \rightarrow (A \rightarrow C)$	S/S	$R \sqsubseteq (RS)/S$	$(T/S)S \sqsubseteq T$	$(S \cup T)R = (SR) \cup (TR)$	$(R \cap R') / S = (R / S) \cap (R' / S)$	$((RS)/S)S = RS$	$((T/S)S)/S = T/S$
10	$S \cdot \dashv S \setminus$	$(B \rightarrow C) \rightarrow (A \rightarrow C)$	$S \setminus S$	$R \sqsubseteq S \setminus (SR)$	$S(S \setminus T) \sqsubseteq T$	$R(S \cup T) = (RS) \cup (RT)$	$S \setminus (R \cap R') = (S \setminus R) \cap (S \setminus R')$	$S(S \setminus (SR)) = SR$	$S \setminus (S(S \setminus T)) = S \setminus T$
11	$R \cap \dashv R \Rightarrow$	$(A \rightarrow B) \rightarrow (A \rightarrow B)$	$X \mapsto R \Rightarrow (X \cap R)$	$X \sqsubseteq R \Rightarrow (X \cap R)$	$R \cap (R \Rightarrow Y) \sqsubseteq Y$	$R \cap (X \cup Y) = (R \cap X) \cup (R \cap Y)$	$R \Rightarrow S \cap T = (R \Rightarrow S) \cap (R \Rightarrow T)$	$R \cap (R \Rightarrow (X \cap R)) = X \cap R$	$R \Rightarrow (R \cap (R \Rightarrow Y)) = R \Rightarrow Y$
12	$\mathcal{D} \dashv \cdot T$	$(A \rightarrow B) \rightarrow \text{Cor } A$	$R \mapsto (\mathcal{D}R)T$	$R \sqsubseteq (\mathcal{D}R)T$	$\mathcal{D}(AT) \sqsubseteq A$	$\mathcal{D}(R \cup S) = \mathcal{D}R \cup \mathcal{D}S$	$(A \cap B)T = AT \cap BT$	$\mathcal{D}((\mathcal{D}R)T) = \mathcal{D}R$	$(\mathcal{D}(AT))T = AT$
13	$\mathcal{R} \dashv T \cdot$	$(A \rightarrow B) \rightarrow \text{Cor } B$	$R \mapsto T(\mathcal{R}R)$	$R \sqsubseteq T(\mathcal{R}R)$	$\mathcal{R}(TA) \sqsubseteq A$	$\mathcal{R}(R \cup S) = \mathcal{R}R \cup \mathcal{R}S$	$T(A \cap B) = TA \cap TB$	$\mathcal{R}(T(\mathcal{R}R)) = \mathcal{R}R$	$T(\mathcal{R}(TA)) = TA$
14	$\cdot f \dashv \cdot f^{\circ}$	$(A \rightarrow B) \rightarrow (A \rightarrow C)$	ff°	$\text{Map}(f) \Rightarrow 1 \sqsubseteq f f^{\circ}$	$f^{\circ} f \sqsubseteq 1$	$(S \cup T)R = (SR) \cup (TR)$	$\text{Simple}(g) \Rightarrow (X \cap Y)g^{\circ} = (Xg^{\circ}) \cap (Yg^{\circ})$	$ff^{\circ}f = f$	$f^{\circ}ff^{\circ} = f^{\circ}$
15	$f^{\circ} \cdot \dashv f \cdot$	$(A \rightarrow C) \rightarrow (B \rightarrow C)$	ff°	$\text{Map}(f) \Rightarrow 1 \sqsubseteq f f^{\circ}$	$f^{\circ} f \sqsubseteq 1$	$R(S \cup T) = (RS) \cup (RT)$	$f(X \cap Y) = fX \cap fY$	$ff^{\circ}f = f$	$f^{\circ}ff^{\circ} = f^{\circ}$
16	$i \dashv E$	$\text{Map} \multimap \text{Rel}$	E	$\frac{1}{\exists} : A \rightarrow PA$	$\exists : PB \rightarrow B$	—	—	$\frac{\exists}{\exists} = 1$	$\frac{R}{\exists} \exists = R$
17	$\cdot \exists \dashv \frac{\exists}{\exists}$	$\text{Map}(A, PB) \rightarrow (A \rightarrow B)$	1	$\frac{f \exists}{\exists} = f$	$\frac{R}{\exists} \exists = R$	—	—	$\frac{\exists}{\exists} \frac{f \exists}{\exists} = f \exists$	$\frac{R}{\exists} \frac{f \exists}{\exists} \exists = f \exists$
18	$\frac{\exists}{\exists} \dashv \cdot \exists$	$(A \rightarrow B) \rightarrow \text{Map}(A, PB)$	1	$\frac{R}{\exists} \exists = R$	$\frac{f \exists}{\exists} = f$	—	—	$\frac{\exists}{\exists} \frac{R}{\exists} = \frac{R}{\exists}$	$\frac{f \exists}{\exists} \frac{R}{\exists} \exists = f \exists$

§0.2. Composing adjunctions

(0.2a)

	$\bullet T$	$T \bullet$	$\bullet g$	$g^\circ \bullet$	Tn	$^\circ$	$\bullet \triangleleft$	Δ
1	$\bullet S$	$R / (S_1 \ S_2)$ $= R / S_2 / S_1$	$S \setminus (R / W)$ $= (S \setminus R) / W$	$R / (Sg) =$ $(Rg^\circ) / S$	$\text{Map}(f) \Rightarrow f$ $(R / S) =$ $(f R) / S$	$(Y/S)^\circ =$ $S^\circ \setminus (Y^\circ)$	$(RS)^\wedge =$ $R^\wedge (S \otimes 1)$	$(R \cap R') /$ $S = (R / S)$ $\cap (R' / S)$
2	$S \bullet$	$S \setminus (R / W)$ $= (S \setminus R) / W$	$(S_1 \ S_2) \setminus R$ $= S_2 \setminus (S_1 \setminus R)$	$S \setminus (Rg^\circ) =$ $(S \setminus R)g^\circ$	$(g^\circ S) \setminus R =$ $S \setminus (gR)$	$(S \setminus Y)^\circ =$ Y° / S°	$((S \otimes 1)R)^\wedge =$ $SR^\wedge$	$S \setminus (R \cap$ $R') = (S \setminus$ $R) \cap (S \setminus$ $R')$
3	$\bullet f$	$\text{Map}(f) \Rightarrow$ $R / (f S) =$ $(R / S) f^\circ$	$T \setminus (Rf^\circ) =$ $(T \setminus R) f^\circ$	$(R S)^\circ = S^\circ$ R°		$(R S)^\circ = S^\circ$ R°		$\text{Simple}(g)$ $\Rightarrow (X \cap Y)$ $g^\circ = (X g^\circ)$ $\cap (Y g^\circ)$
4	$f^\circ \bullet$	$\text{Map}(f) \Rightarrow f$ $(R / S) =$ $(f R) / S$	$(Tf^\circ) \setminus R =$ $f(T \setminus R)$	$(R S)^\circ = S^\circ$ R°		$(R S)^\circ = S^\circ$ R°		$f(T_1 \cap T_2) =$ $fT_1 \cap fT_2$
5	Rn	—	—	—	$R \Rightarrow S \Rightarrow T =$ $R \cap S \Rightarrow T$	$(R \Rightarrow Y)^\circ =$ $R^\circ \Rightarrow (Y^\circ)$	—	$R \Rightarrow S \cap T =$ $(R \Rightarrow S) \cap$ $(R \Rightarrow T)$
6	$^\circ$	$(Y/T)^\circ =$ $T^\circ \setminus (Y^\circ)$	$(T \setminus Y)^\circ =$ Y° / T°	$(R S)^\circ = S^\circ$ R°	$(R S)^\circ = S^\circ$ R°	$(T \Rightarrow Y)^\circ =$ $T^\circ \Rightarrow (Y^\circ)$	$R^\circ \circ = R$	$(T_1 \cap T_2)^\circ =$ $T_1^\circ \cap T_2^\circ$
7	$\bullet \triangleleft$	$(RT)^\wedge =$ $R^\wedge (T \otimes 1)$	$((T \otimes 1)R)^\wedge =$ $TR^\wedge$	—	—	$(R^\wedge)^\circ =$ $(R^\circ \otimes 1)$ $(1 \otimes \triangleright \bullet)$ both bends: the $^\circ$ section's display	—	—
8	u	$(S \cup T) R =$ $(S R) \cup (T R)$	$R (S \cup T) =$ $(R S) \cup (R T)$	$(S \cup T) R =$ $(S R) \cup (T R)$	$R (S \cup T) =$ $(R S) \cup (R T)$	$R \cap (S \cup T) =$ $R \cap S \cup R \cap T$	$(R \cup S)^\circ =$ $S^\circ \cup R^\circ$	
9	$\emptyset$	$\emptyset R = \emptyset$	$R \emptyset = \emptyset$	$\emptyset R = \emptyset$	$R \emptyset = \emptyset$	$R \cap \emptyset = \emptyset$	$\emptyset^\circ = \emptyset$	

§0.3. Adjoint triples

(0.3a)

$$\bullet f \dashv \bullet f^\circ \dashv / f^\circ$$

$$f^\circ \bullet \dashv f \bullet \dashv f \setminus$$

§0.4. Frobenius

Frobenius.

(0.4a)

$$(n \otimes n) \otimes n = n \otimes (n \otimes n)$$

$\triangleright$ associative

Frobenius, one half — the other is its $^\circ$

$$n \otimes n = n \otimes n$$

$\triangleright$ commutative

$$R \triangleleft \leq \triangleleft (R \otimes R)$$

$$n \otimes I = n$$

$\bullet$ is its unit

$$R \multimap \leq \multimap$$

$R \multimap \leq \multimap$

(0.2a)

$$\begin{array}{cccc}
 \begin{array}{c} n \otimes n \\ \triangleleft \triangleright \\ n \otimes n \end{array} & \leq & \begin{array}{c} n \otimes n \\ \triangleright \triangleleft \\ n \otimes n \end{array} & \begin{array}{c} I \\ \triangleleft \triangleright \\ I \end{array} \\
 \triangleleft \triangleright \leq 1 & & 1 \leq \triangleright \triangleleft & \begin{array}{c} I \\ \triangleright \triangleleft \\ I \end{array} \\
 & & & \begin{array}{c} I \\ \triangleright \triangleleft \\ I \end{array} \\
 & & & \begin{array}{c} I \\ \triangleright \triangleleft \\ I \end{array}
 \end{array}
 \quad (0.4b)$$

§0.5. $\circ$ is a contravariant 2-functor

(0.5a)

$\circ$ is primitive, part of the data the first section lists. It turns both of R 's wires round, and the Frobenius structure **DRAWS** that — the picture and the formula below are R° , not its definition:



$$R^\circ = (\circ \triangleleft \otimes 1) (1 \otimes R \otimes 1) (1 \otimes \triangleright \circ)$$

where $\circ \triangleleft : I \rightarrow a \otimes a$ opens a pair of wires out of nothing and $\triangleright \circ : b \otimes b \rightarrow I$ closes one, so the input of R° is where the output of R was. Taking that formula as the definition would now be circular: $\triangleleft$ and $\circ$ are $\triangleright^\circ$ and $\circ^\circ$. The laws of $\circ$ are that it is a contravariant 2-functor $\circ : \mathcal{C}^{\circ p} \rightarrow \mathcal{C}$:

$$(i) \ 1^\circ = 1 \quad (ii) \ (RS)^\circ = S^\circ R^\circ \quad (iii) \ (R \otimes S)^\circ = R^\circ \otimes S^\circ \quad (iv) \ R \leq S \text{ implies } R^\circ \leq S^\circ$$

§0.6. Meet and composition

(0.6a)

semi-distributivity, and what supplies it	picture
$R \ (S \cap T) \sqsubseteq RS \cap RT$ — the lax copy law. Equality exactly when R is single valued: the Maps section's $F(R \cap S) = FR \cap FS$.	

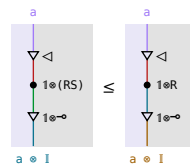
§0.7. Domain and range

(0.7a)

1	$\text{dom}(R) \triangleq 1 \cap (R \ R^\circ)$
2	$\text{Coreflexive}(X) \implies \text{dom}(R) \sqsubseteq X \iff R \sqsubseteq X \ R$
3	$\text{dom}(R \ S) \sqsubseteq \text{dom}(R)$
4	$\text{dom}(R \cap S) = 1 \cap (S \ R^\circ)$
5	$R \text{ entire } \text{Entire}(R) \triangleq \text{dom}(R) = 1$
6	$R \text{ simple } \text{Simple}(R) \triangleq R^\circ \ R \sqsubseteq 1$
7	$R \text{ a map } \text{Map}(R) \triangleq \text{Entire}(R) \wedge \text{Simple}(R)$
8	$\text{Entire}(R) \wedge \text{Entire}(S) \implies \text{Entire}(R \ S)$ $\text{Simple}(f) \wedge \text{Simple}(g) \implies \text{Simple}(f \ g)$ $\text{Map}(f) \wedge \text{Map}(g) \implies \text{Map}(f \ g)$
9	$\text{Entire}(R \ S) \implies \text{Entire}(R)$

§0.8. Sliding the discard

(0.8a)



$S \multimap \multimap \multimap$, the lax axiom for $\multimap$ in the first section
— the discard slides back past S

§0.9. Division

(0.9a)

1	$X \in R/S \iff X \in R$ X is any X -to- Y pairing; one that only pairs X with a Y such that X admires everyone Y hates lies inside R/S , and R/S is the largest such.	
2	$X \in S \setminus R \iff S \setminus X \in R$ The mirror — divide on the left when X comes first.	
3	$(R/S) \in R$ There is a Y such that X admires everyone Y hates, and p is one of the people Y hates — then X admires p too. Strict at $S=\emptyset$: R/S is everyone, $(R/S)S=\emptyset$.	
4	$S \setminus (S \setminus R) \in R$ The mirror.	
5	associate: $R/(S_1 S_2) = (R/S_2)/S_1$ A friend's enemy is two hops: divide by the far end first.	
6	$(S_1 S_2) \setminus R = S_2 \setminus (S_1 \setminus R)$ The mirror.	
7	maps: $f \cdot (R/S) = (fR)/S$ Rename X before or after dividing — the licence to write fR/S .	
8	$R/(fS) = (R/S) f^\circ$ Rename Y : a map leaves a denominator as f° outside the box.	
9	$(R/S) \cdot (S/W) \in R/W$ Someone who admires all of a hate-set that already covers everyone Z works for admires those people too.	

(0.8a)

10	$1 \sqsubseteq R/R$ R/R runs admirer to admirer: each admires everyone they admire. Strict: two people who each admire only a and b admire each other's idols too, and still stay two people.	
11	$(R/R) (R/R) = R/R$ R/R is the preorder admires at least as much as, and a preorder is idempotent. Freyd writes $\sqsubseteq$; with $1 \sqsubseteq R/R$ above it is an equality.	
12	$(R/R)R = R$ Reaching p through someone whose idols x fully admires is reaching p directly, since x admires their own idols.	
13	$R/1 = R$ Dividing by 1: p's set is just $\{p\}$, so admiring all of it is admiring p.	
14	$R/(S_1 \cup S_2) = R/S_1 \cap R/S_2$ Admiring a combined hate-set is admiring each set in full.	
15	$S \setminus (R/W) = (S \setminus R)/W$ Which is why $S \setminus R/W$ needs no bracket.	

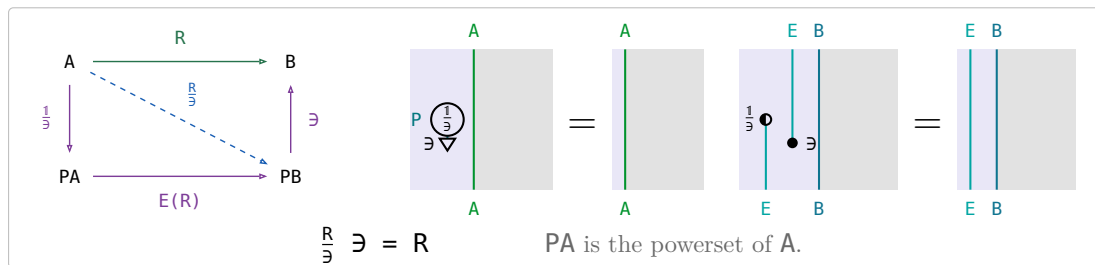
§0.10. Power allegories

(0.10a)

	law	the reading
1	$\exists_R \square = R\square, \exists_R = \exists_R \square$	One $\exists$ per object, not per arrow.
2	$\exists$ is thick	Comprehension: every x has a set of exactly the people x admires.
3	$\frac{R}{\exists} : A \rightarrow PB$, for $R : A \rightarrow B$	convert a relation to a function. $a \frac{R}{\exists} = \{b \mid a R b\}$
4	$\text{Map}(\frac{R}{\exists})$	
5	$\frac{R}{\exists} \exists = R$	reading the chosen set back through $\exists$ returns the relation.
6	$\text{Simple}(F) \Rightarrow F \sqsubseteq \frac{F \exists}{\exists}$	A partial choice of sets is inside the total one.
7	the singleton map is monic: $\frac{1}{\exists} (\frac{1}{\exists})^\circ \sqsubseteq 1$ two points with the same one-person set are the same point.	The one-person set.
8	$\frac{\exists}{\exists} = 1$	Make the set of a set, then read it back one level down.
9	fusion: $\text{Map}(f) \Rightarrow \frac{f R}{\exists} = f \frac{R}{\exists}$	Naturality of the unit, $\text{Map}(f) \Rightarrow f \frac{1}{\exists} = \frac{1}{\exists} E(f)$. renaming a point and then taking its one-person set is taking the set and renaming inside it.
10	$\text{Map}(f) \wedge 1\square = \exists\square \Rightarrow \frac{f}{\exists} = f \frac{1}{\exists}$	Rename first or take singletons first — the fusion row above at $R=1$.
11	$R\square = \exists\square \Rightarrow R /_s S = \frac{R}{\exists} (\frac{S}{\exists})^\circ$	x and y match when R sends x and S sends y to the same set.
12	$E(R) \triangleq \frac{\exists R}{\exists}$	$E(R) : PA \rightarrow PB$, image of a set of A
13	$\frac{R}{\exists} = \frac{1}{\exists} E(R)$	$\frac{1}{\exists} : x \mapsto \{x\}$
14	$\frac{S}{\exists} E(R) = \frac{S R}{\exists}$	absorption — the monad's composition law, $\frac{S}{\exists} \diamond \frac{R}{\exists} = \frac{S R}{\exists}$, $\mathcal{A} \cong \text{Kleisli}(E)$ in Relation Algebra
15	$E(R) = E(\frac{R}{\exists})$ union	the image of a set under R is the union of the R -sets of its points.
16	$\supseteq \triangleq \exists / \exists$	$xs \supseteq ys \Leftrightarrow \forall a. ys \exists a \rightarrow xs \exists a$
17	$\subseteq = \supseteq^\circ$	$xs \subseteq ys \Leftrightarrow ys \supseteq xs$
18	$T^\circ \frac{T}{\exists} \sqsubseteq E$	every T -value of a point is a member of its T -set: $\wedge(T)$ is a map and $\wedge(T)\exists=T$.

§0.11. $\mathbf{i}\text{-E}$: power transpose through singleton

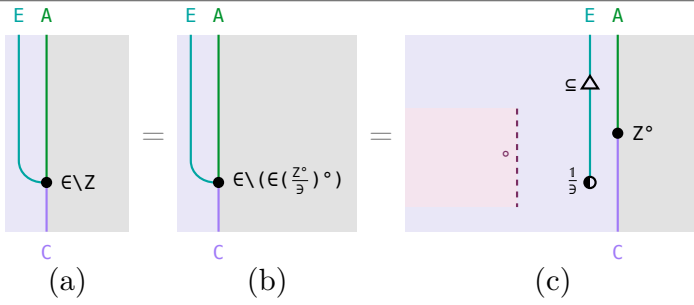
(0.11a)



§0.12. $\epsilon \setminus$ as a composite

$$\epsilon \setminus Z = \subseteq \left(\frac{Z^\circ}{3} \right)^\circ \quad (0.12a)$$

$\mathbf{x}\mathbf{s}$ is related to $\mathbf{c}$ by $\epsilon \setminus Z$ exactly when $\mathbf{x}\mathbf{s} \subseteq \Lambda(Z^\circ)(\mathbf{c})$, the set of every $\mathbf{a}$ with $\mathbf{a} Z \mathbf{c}$



(a)	$\epsilon \setminus Z$	$Z = \epsilon \Lambda(Z^\circ)^\circ \text{ — (0.10a.5)}$
(b)	$= \epsilon \setminus \left(\epsilon \setminus \left(\frac{Z^\circ}{3} \right)^\circ \right)$	$X \setminus (Y f^\circ) = (X \setminus Y) f^\circ, f \text{ a map}$
(c)	$= \subseteq \left(\frac{Z^\circ}{3} \right)^\circ$	